
EXERCISES

Homework 4

AI503: Calculus

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1 Problem 1

Consider $f(x, y) = x^2 + y^2 - 4x - 6y$.

1. Find the critical points of f .
2. Classify them (minimum, maximum, or saddle point) using the Hessian.

1.1 Solution

1. We first find the derivatives

$$f_{x(x,y)} = 2x - 4$$

$$f_{y(x,y)} = 2y - 6$$

We find the critical point(s) at

$$f_{x(x,y)} = 0$$

$$2x - 4 = 0 \Rightarrow x = \frac{4}{2} = 2$$

$$f_{y(x,y)} = 0$$

$$2y - 6 = 0 \Rightarrow y = \frac{6}{2} = 3$$

f only has 1 critical point on $(2, 3)$

2. We know the hessian as

$$H(f)(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial xy} \\ \frac{\partial^2 f}{\partial yx} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

this results to

$$H(f)(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Since we know that at the critical point

$$\det \left(\begin{bmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \right) = 0$$

We can then find that: $(2 - \lambda) \cdot (2 - \lambda) = 0$ and $\lambda = 2$

This is then a local minimum by

Second-Derivative Test Hessian

At a critical point P_0 , let $H = H(f)(P_0)$ be the Hessian matrix: Compute eigenvalues of H .

- All positive eigenvalues $\Rightarrow$ local minimum
- All negative eigenvalues $\Rightarrow$ local maximum
- Mixed signs $\Rightarrow$ saddle point
- Zero eigenvalue(s) $\Rightarrow$ test inconclusive

2 Problem 2

Let $f(u, v) = u^2 + 3uv$, $u(x, y) = x^2 - y$, and $v(x, y) = \sin(xy)$.

Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ using the multivariable chain rule.

2.1 Solution

For this we use the multivariate chain rule for multivariable functions

Multivariate Chain Rule

For $f(g, h)$, with $g(x, y)$ and $h(u, v)$, then

$$\begin{aligned} \bullet \quad \frac{\partial f}{\partial x} &= \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} \\ \bullet \quad \frac{\partial f}{\partial y} &= \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} \end{aligned}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x}$$

$$\begin{aligned} \frac{\partial f}{\partial x} &= (2u + 3v) \cdot 2x + 3u \cdot (\cos(xy) \cdot y) \\ &= 4xu + 6xv + 3yu \cdot \cos(xy) \\ &= 4x(x^2 - y) + 6x(\sin(xy)) + 3y(x^2 - y) \cdot \cos(xy) \\ &= 4x^3 - 4xy + 6x \sin(xy) + 3yx^2 \cos(xy) - 3y^2 \cos(xy) \end{aligned}$$

We also know that

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y}$$

So.. again

$$\begin{aligned} \frac{\partial f}{\partial y} &= (2u + 3v) \cdot (-1) + 3u \cdot (\cos(xy) \cdot x) \\ &= -2u - 3v + 3xu \cdot \cos(xy) \\ &= -2(x^2 - y) - 3 \cdot (\sin(xy)) + 3x(x^2 - y) \cdot \cos(xy) \\ &= -2x^2 + 2y - 3 \cdot \sin(xy) + 3x^3 \cdot \cos(xy) - 3xy \cdot \cos(xy) \end{aligned}$$

So finally

$$\frac{\partial f}{\partial x} = 4x^3 - 4xy + 6x \sin(xy) + 3yx^2 \cos(xy) - 3y^2 \cos(xy)$$

$$\frac{\partial f}{\partial y} = -2x^2 + 2y - 3 \cdot \sin(xy) + 3x^3 \cdot \cos(xy) - 3xy \cdot \cos(xy)$$

3 Problem 3

A rectangular box with square base and volume $V = 1\text{m}^3$ is to be built with the least surface area.

- (a) Express the surface area $S(x, h)$ in terms of base side length x and height h .
- (b) Eliminate one variable using the volume constraint.
- (c) Use calculus to find the dimensions minimizing S .

3.1 Solution

- (a) Since the box must have a base of x^2 (top and bottom) as well as 4 sides of $4xh$

$$S(x, h) = 2x^2 + 4xh$$

- (b) Since volume is constrained, we know that $1\text{m}^3 = x^2h$

because of this

$$h = \frac{1}{x^2}$$

then substitute h in the function for $S(x, h)$

this gives

$$S(x) = 2x^2 + \frac{4}{x}$$

- (c) Take the derivative of $S(x)$ and solve for $\frac{\partial S}{\partial x} S(x) = 0$

$$\frac{\partial S}{\partial x} S(x) = 4x - 4x^{-2} = 4x - \frac{4}{x^2}$$

$$0 = 4x - \frac{4}{x^2}$$

$$\frac{4}{4} = x^3 = 1$$

S must have an critical point on $x = 1$

Now to find h

$$h = \frac{1}{(1)^2} = 1$$

To conclude. $x = 1$ and $h = 1$

Note that

$$S(1) = 2(1)^2 + \frac{4}{1^2} = 2 + 4 = 6\text{m}^2$$

4 Problem 4

Two products are manufactured in quantities q_1 and q_2 and sold at prices p_1 and p_2 , respectively. The cost of producing them is given by

$$C = 2q_1^2 + 2q_2^2 + 10$$

- (a) Find the maximum profit that can be made, assuming the prices are fixed.
- (b) Find the rate of change of that maximum profit as p_1 increases.

4.1 Solution

1. To maximize profit, we should first define the profit function

Profit must be

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

$$\pi(q_1, q_2) = R(q_1, q_2) - C(q_1, q_2)$$

$$\pi(q_1, q_2) = p_1 q_1 + p_2 q_2 - (2q_1^2 + 2q_2^2 + 10)$$

$$\pi(q_1, q_2) = p_1 q_1 + p_2 q_2 - 2q_1^2 - 2q_2^2 - 10$$

$$\pi(q_1, q_2) = -2q_1^2 + p_1 q_1 - 2q_2^2 + p_2 q_2 - 10$$

Now find $\frac{\partial \pi}{\partial q_1}$ and $\frac{\partial \pi}{\partial q_2}$

$$\frac{\partial \pi}{\partial q_1} = -4q_1 + p_1 \iff q_1 = \frac{p_1}{4}$$

$$\frac{\partial \pi}{\partial q_2} = -4q_2 + p_2 \iff q_2 = \frac{p_2}{4}$$

These are the profit-maximizing quantities

So...

$$\begin{aligned}\pi_{\max} &= P\left(\frac{p_1}{4}, \frac{p_2}{4}\right) = -2\left(\frac{p_1}{4}\right)^2 + p_1\left(\frac{p_1}{4}\right) - 2\left(\frac{p_2}{4}\right)^2 + p_2\left(\frac{p_2}{4}\right) - 10 \\ &= -\frac{p_1^2}{8} + \frac{p_1^2}{4} - \frac{p_2^2}{8} + \frac{p_2^2}{4} + 10 \\ &= \frac{p_1^2}{8} + \frac{p_2^2}{8} + 10\end{aligned}$$

5 Problem 5

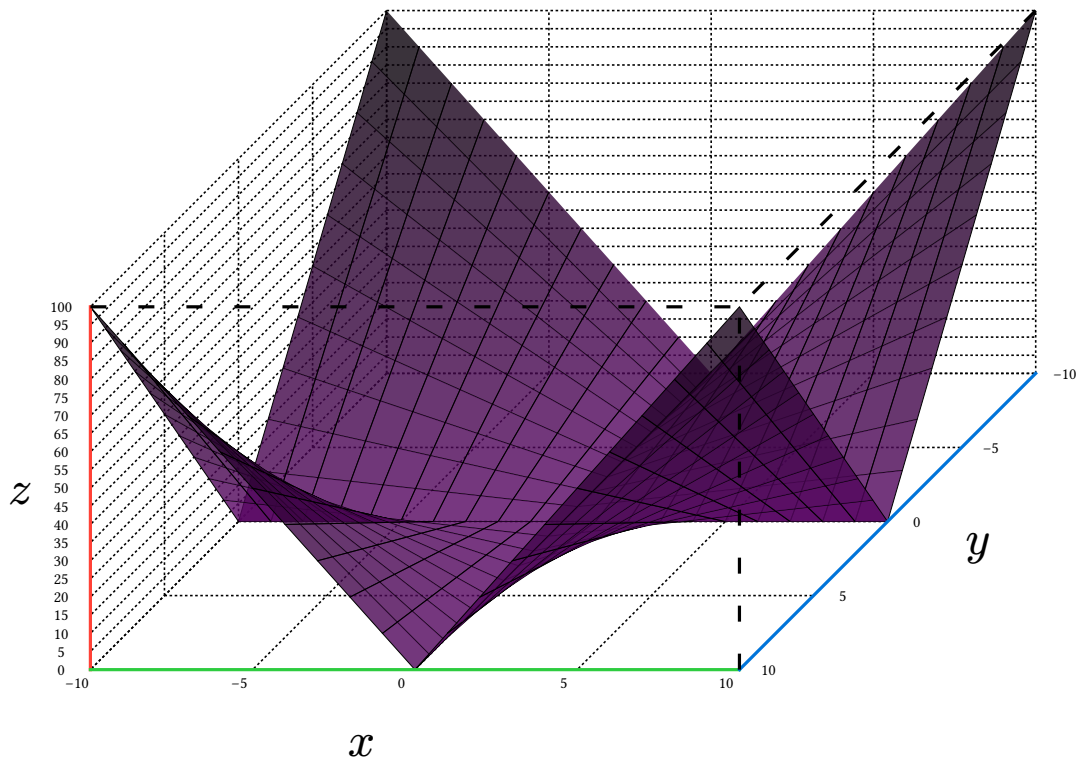
Consider the function $f(x, y) = |xy|$.

1. Use a computer to draw the graph of f . Does the graph look like a plane when we zoom in on the origin?
2. Is f differentiable at $(x, y) \neq (0, 0)$?
3. Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist.
4. Are f_x and f_y continuous at $(0, 0)$?
5. Is f differentiable at $(0, 0)$?

Hint: Consider the directional derivative $f_u(0, 0)$ for $u = (i + j)(\sqrt{2})$.

5.1 Solution

1. The graph of $f(x, y) = |xy|$:



When we zoom in on the origin, the graph does NOT look like a plane - it has sharp ridges along the coordinate axes due to the absolute value function.

2. Input $(0, 0)$ into the $f_x(x, y)$ and $f_y(x, y)$

$$\begin{aligned}f_x(0, 0) &= \lim_{\varepsilon \rightarrow 0} \frac{f(\varepsilon, 0) - f(0, 0)}{\varepsilon} \\&= \lim_{\varepsilon \rightarrow 0} \frac{0 - 0}{\varepsilon} = \lim_{\varepsilon \rightarrow 0^+} \frac{0}{\varepsilon} = \lim_{\varepsilon \rightarrow 0^-} \frac{0}{\varepsilon} = 0 \\f_y(0, 0) &= \lim_{\varepsilon \rightarrow 0} \frac{f(0, \varepsilon) - f(0, 0)}{\varepsilon} \\&= \lim_{\varepsilon \rightarrow 0} \frac{0 - 0}{\varepsilon} = \lim_{\varepsilon \rightarrow 0^+} \frac{0}{\varepsilon} = \lim_{\varepsilon \rightarrow 0^-} \frac{0}{\varepsilon} = 0\end{aligned}$$

They both exists

3. First find f_x and f_y

$$\begin{aligned}\frac{\partial}{\partial x} f(x, y) &= \frac{\partial}{\partial x} |xy| = |y| \frac{\partial}{\partial x} |x| = |y| \operatorname{sgn}(x) \\ \frac{\partial}{\partial y} f(x, y) &= \frac{\partial}{\partial y} |xy| = |x| \frac{\partial}{\partial y} |y| = |x| \operatorname{sgn}(y)\end{aligned}$$

where

$$\operatorname{sgn}(x) = \begin{cases} +1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

So now check for $(x, y) \rightarrow (0, 0)$

$$\lim_{(x, y) \rightarrow (0, 0)} f_x(x, y) = |0| + \operatorname{sgn}(0) = 0$$

$$\lim_{(x, y) \rightarrow (0, 0)} f_y(x, y) = |0| + \operatorname{sgn}(0) = 0$$

So both exists

4. Since

$$\lim_{(x, y) \rightarrow (0, 0)} f_x(x, y) = 0 = f_x(0, 0)$$

$$\lim_{(x, y) \rightarrow (0, 0)} f_y(x, y) = 0 = f_y(0, 0)$$

They are both continuous at $(0, 0)$

5. For the directional derivative $f_u(0, 0)$ for $\vec{u} = \frac{i+j}{\sqrt{2}} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$.

$$\begin{aligned} f_u(x, y) &= \nabla f(x, y) \cdot \vec{u} = (f_x(x, y), f_y(x, y)) \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \\ &= f_x(x, y) \cdot \frac{1}{\sqrt{2}} + f_y(x, y) \cdot \frac{1}{\sqrt{2}} \\ &= \frac{|y| \cdot \operatorname{sgn}(x)}{\sqrt{2}} + \frac{|x| \cdot \operatorname{sgn}(y)}{\sqrt{2}} \\ &= \frac{|y| \cdot \operatorname{sgn}(x) + |x| \cdot \operatorname{sgn}(y)}{\sqrt{2}} \\ f_u(0, 0) &= \frac{|0| \cdot \operatorname{sgn}(0) + |0| \cdot \operatorname{sgn}(0)}{\sqrt{2}} = 0 \end{aligned}$$

So it is differentiable at $(0, 0)$ with $f_u(0, 0) = \mathbf{0}$